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College of Engineering - Department of Electronics Engineering

COLLEGE OF ENGINEERING
Department of Electronics Engineering
BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING

ECE 410 - ELECTRONICS SYSTEM AND DESIGN 2

TECHCULTIVATE:
BRIDGING NATURE AND INNOVATION FOR SUSTAINABLE GROWTH
A FINAL PROJECT

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ECE - 3205

Submitted to:
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May 12, 2024



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I. INTRODUCTION

Optimizing plant growth and productivity amidst climate change challenges is crucial in today's agricultural landscape. Controlled climate environment systems in greenhouses offer a solution by regulating temperature, humidity, light, and air quality to create ideal growing conditions while shielding crops from adverse weather and environmental stresses. Leveraging IoT technology for remote monitoring and control and automation enhances efficiency, maximizing yields while conserving resources and promoting sustainability. This study aims to develop a system that effectively controls greenhouse environmental factors, shaping the future of agriculture towards greater efficiency, productivity, and environmental stewardship.

II. STATEMENT OF THE PROBLEM

The conventional greenhouse management methods lack precision, real-time feedback, and adaptive capabilities, resulting in inefficiencies, resource wastage, and reduced productivity. Integrating IoT technology into greenhouse systems promises to address these limitations by enabling remote monitoring of environmental parameters such as temperature, humidity, soil moisture, and light intensity. Successful implementation of IoT in greenhouse operations requires overcoming various technical, economic, and logistical challenges. These challenges include the selection of appropriate IoT sensors and devices, the establishment of reliable connectivity infrastructure, the integration of disparate data sources, and the development of user-friendly interfaces for data interpretation and control.

III. OBJECTIVES

This project aims to:

1. To develop an automated system platform with sensors and actuators to monitor the greenhouse environment such as temperature, moisture, and rainfall.
2. To integrate rain sensors into the system to detect rainfall and automatically close the greenhouse roof to protect plants from excessive water exposure and prevent waterlogging.
3. To prioritize efficiency and safety in climate control by optimizing ventilation, minimizing the risk of overheating, and safeguarding the plant health.



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IV. SIGNIFICANCE OF THE PROJECT

Understanding the potential benefits, challenges, and implications of integrating IoT technology into greenhouse operations is essential for advancing agricultural sustainability, enhancing food security, and mitigating the environmental impact of conventional farming practices. By investigating the technical, economic, and logistical aspects of IoT integration in greenhouses, this study aims to provide insights and recommendations to stakeholders in the agriculture sector, including growers, policymakers, researchers, and technology providers, to facilitate informed decision-making and promote the adoption of IoT-enabled solutions for sustainable agriculture.

V. METHODOLOGY

The group embarked on their goal by meticulously following the block diagram, starting with hand sketching to visualize the idea of the greenhouse. Selecting the sensors and actuators to be used in the system, testing various algorithms (programs), and implementing them into the MCU. Constructing the structure using 3D Models or workshop tools and seamlessly merging all components into a functional system.

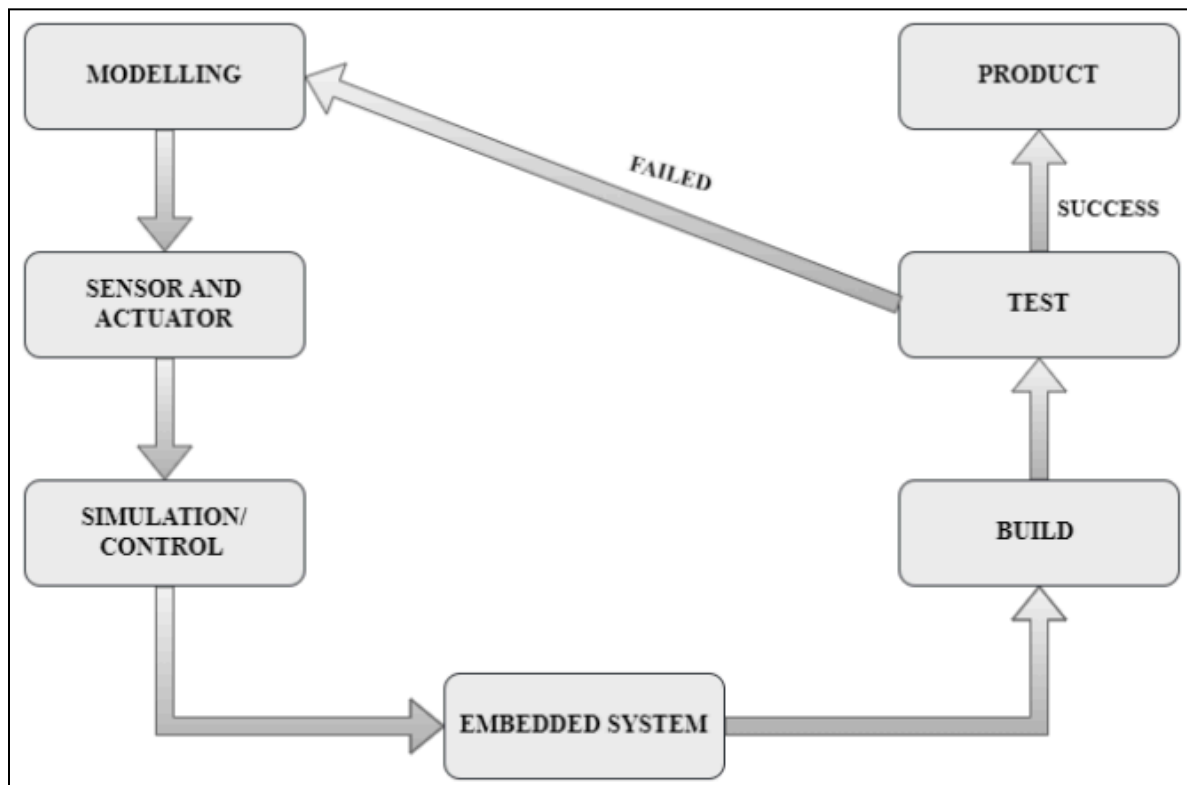


Figure 1. Block diagram of the Proposed System



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The flowchart illustrates a cyclical process in the greenhouse system, where parameters are read, data is transmitted, and the cycle repeats after storing data in Arduino Cloud/ BLYNK IoT.

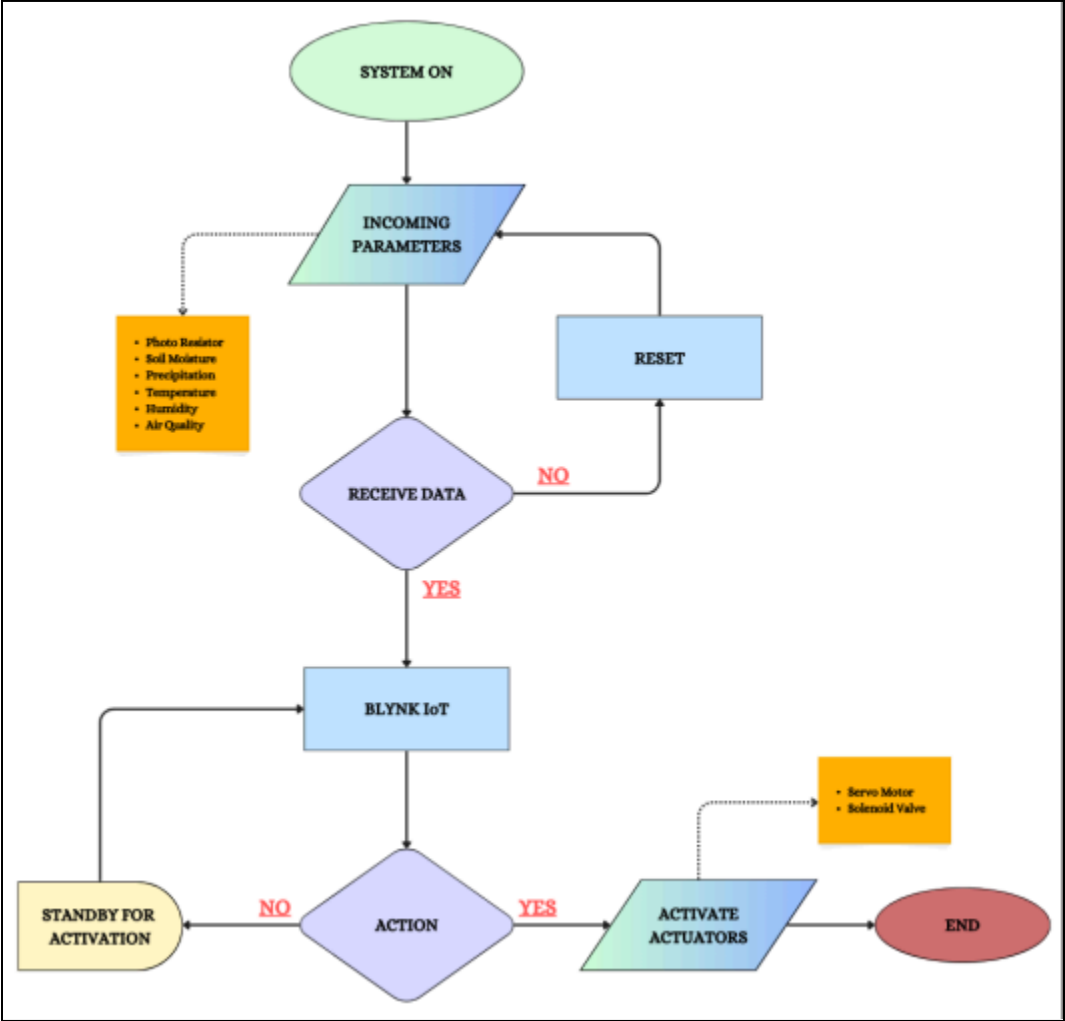


Figure 2. Flowchart of the Proposed System

The process begins with receiving incoming parameters. If data is received it proceeds to Arduino Cloud/ BLYNK IoT to be accessible to the user. However, if the incoming data is not received, the system resets and repeats until the data is successfully received. Then from the User Interface, it'll be the wireless control to activate the actuators.



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VI. MATERIALS AND PLAN

In this section, the students outline the key components essential for constructing and operating the innovative greenhouse. It also provides a brief overview of the chosen materials for the greenhouse system.

Input System	
1. Rain Sensor	Measures the precipitation and when it says raining it will automatically close the ventilation.
2. Moisture Sensor	Measures the moisture content in soil and data will be showed in dashboard
3. Photoresistor	A light dependent resistor whose resistance changes based on the intensity of incident light.
4. BME680	Measures Temperature, Humidity and AirQuality to help properly control the microclimate inside the greenhouse.
Output System	
1. Servo Motor	Will be manually controlled through a dashboard to control the temperature inside the greenhouse.
2. IR speed sensor	For a smart curtain that will limit the sun rays passing through the greenhouse it will stop the servo motor to stop rotating.
3. Solenoid Valve	Will be manually controlled through a dashboard for water to pass through and water the crops.
4. LEDs	Is for the lighting system of the LED and will be manually open when night comes.
Power System	
1. Solar Panel	Primary Source of Power
2. Li-Po Battery	Secondary Source of Power
3. Power Switch	For the battery
4. Voltage Sensor	To monitor the voltage from solar panel
5. Power Relay	Will decide to let the solar panel power the system based on the voltage read by the voltage sensor.
6. Power Failure	To switch the power source between Solar Panel and Battery.
7. Voltage Regulator	To regulate the voltage flowing through the ESP32 to maintain sustainability.

Table 1. Components of the Proposed System



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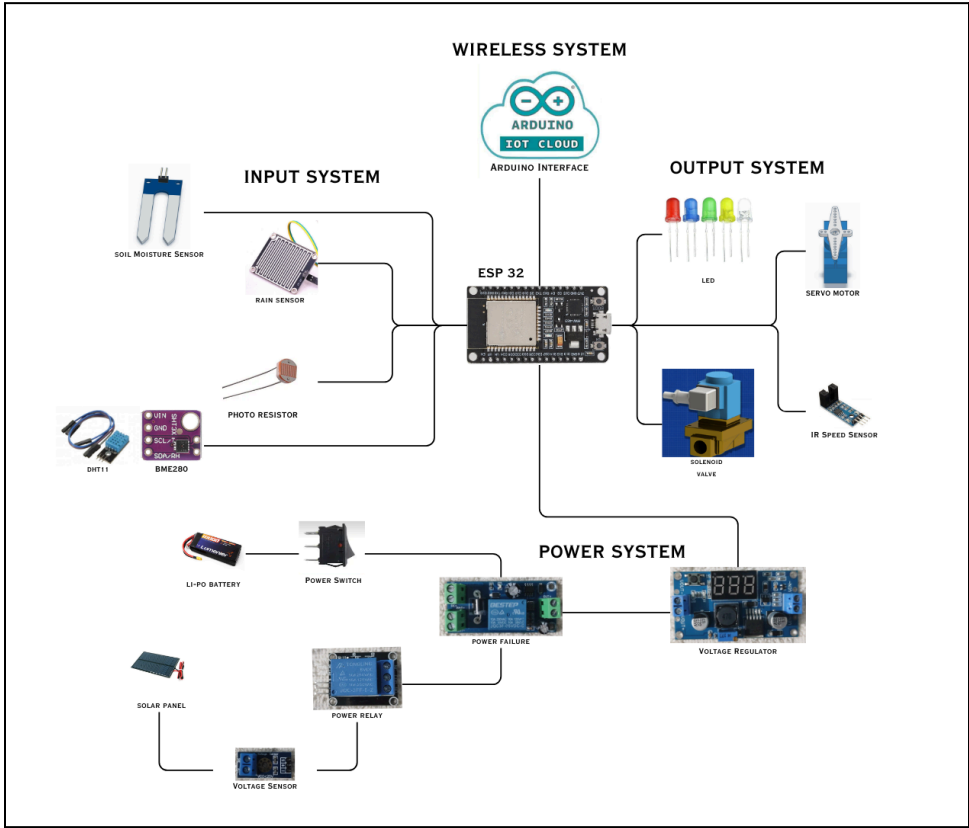


Figure 3. System Architecture of the Proposed System

The figure shows the system architecture of the desired project, illustrating how the group will achieve the integration of technology systems into the greenhouse environment for better understanding.



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VII. PROTOTYPE MAKING



Figure 4. Assembling the Enclosure of the Greenhouse

Figure 4 shows the assembling of the Greenhouse's enclosure. The students build an ideal size of a Greenhouse to be able to test and measure the parameters of the components, specifically the Servo Motor, IR Speed Sensor, Solenoid Valve, and LEDs in the output system. In addition, the Rain Sensors, Moisture Sensors, Photoresistor, and BME680 are for the input system and also, for the power system such as the Solar Panel, Li-Po Battery, Power Switch, Voltage Sensor, Power Relay, Power Failure, and Voltage Regulator. These components are very important in building a greenhouse



Figure 4.1 PCB Making of the System



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Figure 4.1 shown above shows the PCB making of the system. From the first procedure of ironing the printed circuit, taking off the printer circuit in the PCB, applying a ferric chloride, drilling the pins, soldering the pins of the ESP32 microcontroller, and checking the connectivity of the circuit.



Figure 4.2 Connecting the wires on the Enclosure and Microcontroller

Figure 4.2 shows the connecting of the wires on the enclosure and to the ESP32 Microcontroller. Before connecting the wires, the students test first the connectivity and functionality of the sensors. After testing it, the students proceed to connect it to the wires.



Figure 4.3 Designing the Enclosure



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Figure 4.3 above displays the design of the Enclosure. Designing the interior of the enclosure will make the Greenhouse more pleasing to the students. The students decided to add fake grass and pebbles together with the LEDs, solenoid valve, and moisture sensor inside of the enclosure.



Figure 4.4 Finished Output of the Greenhouse

Figure 4.4 shows the finished output of the Greenhouse. The students finally finished connecting the wiring and designing and polishing the enclosure. In addition, the students also test the sensors to make sure that all of them are working properly. Also, the students checked if the servo motor is working before designing the interior just to avoid ruining the grass and pebbles. With these, the students can finally now present their proposed project.



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VIII. CODING

A. EASY EDA

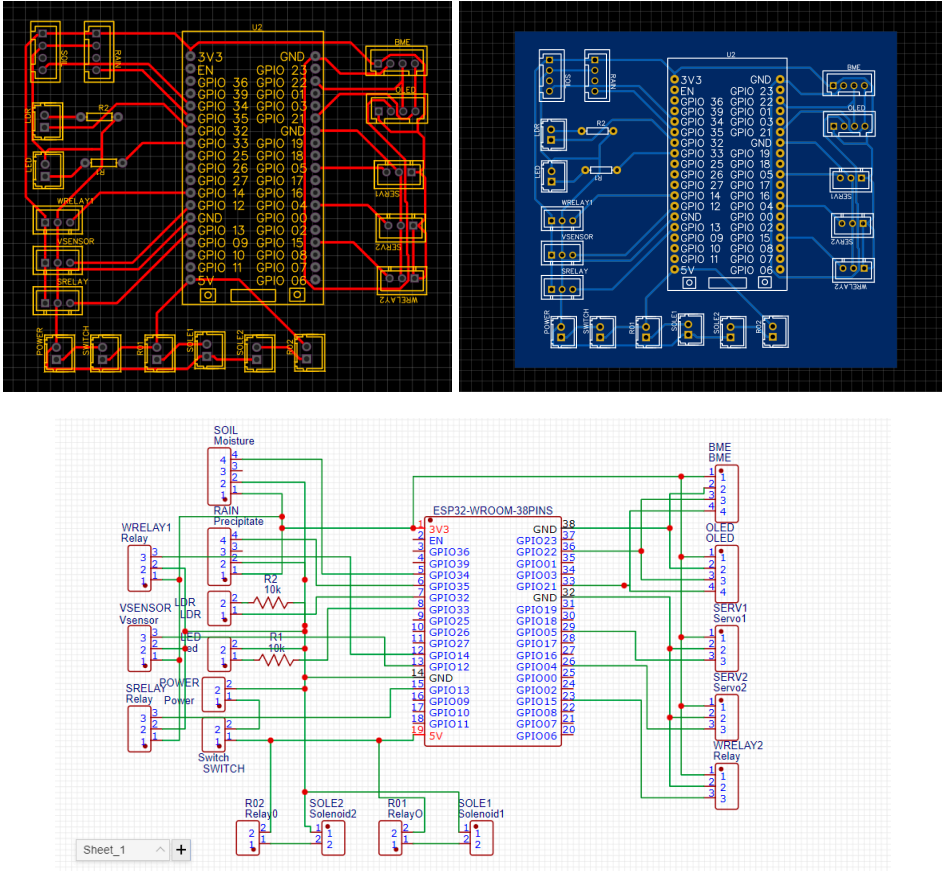


Figure 5. Schematic Design of the Circuit using Easy EDA



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B. ARDUINO IDE

```
1  #include <Wire.h>
2  #include <SPI.h>
3  #include <Adafruit_Sensor.h>
4  #include "Adafruit_BME680.h"
5  #include <ESP32Servo.h>
6
7  #define BLYNK_TEMPLATE_ID "TMPL64AcIzWbK"
8  #define BLYNK_TEMPLATE_NAME "TechCultivate"
9  #define BLYNK_AUTH_TOKEN "0c7Pp72T-4RpFVN9GqYO_haROCKw7EgU"
10
11 #define BLYNK_PRINT Serial
12 #include <WiFi.h>
13 #include <BlynkSimpleEsp32.h>
14
15 char auth[] = BLYNK_AUTH_TOKEN;
16 char ssid[] = "Smart_Bro_0E109";
17 char pass[] = "DioSteBendiga08";
18
19 Adafruit_BME680 bme;
20
21 #define precipita 35 //rain
22 #define moisture 34 //soil
23 bool isRaining = false;
24
25 const int photoResistorPin = 32; //automatic light
26 const int ledPin = 23;          //LED light
27
28 int RelayPin1 = 14;
29 int RelayPin2 = 15;
30
31 const int voltageSensor = 12;
32 float vOUT = 0.0;
33 float vIN = 0.0;
34 float R1 = 30000.0;
35 float R2 = 7500.0;
36 int value = 0;
37 int RelayPin = 13;
38
39 Servo servo1;
40 Servo servo2;
41
42 void setup() {
43   Serial.begin(9600);
```



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```
44 Blynk.begin(auth, ssid, pass);
45 delay(1000);
46 Wire.begin();
47
48 while (!Serial);
49 Serial.println(F("BME680 async test"));
50
51 if (!bme.begin()) {
52     Serial.println(F("Could not find a valid BME680 sensor, check wiring!"));
53     while (1);
54 }
55
56 // Set up oversampling and filter initialization
57 bme.setTemperatureOversampling(BME680_OS_8X);
58 bme.setHumidityOversampling(BME680_OS_2X);
59 bme.setPressureOversampling(BME680_OS_4X);
60 bme.setIIRFilterSize(BME680_FILTER_SIZE_3);
61 bme.setGasHeater(320, 150); // 320°C for 150 ms
62
63 delay(1000);
64
65 pinMode(precipita, INPUT);
66 pinMode(moisture, INPUT);
67 pinMode(ledPin, OUTPUT);
68
69 servo1.setPeriodHertz(50); // Standard 50 Hz servo
70 servo2.setPeriodHertz(50); // Standard 50 Hz servo
71 servo1.attach(5, 500, 2400); // Attach servo 1 to pin 2, with min and max pulse width
72 servo2.attach(4, 500, 2400); // Attach servo 2 to pin 4, with min and max pulse width
73
74 pinMode(RelayPin1, OUTPUT);
75 pinMode(RelayPin2, OUTPUT);
76
77 pinMode(RelayPin, OUTPUT);
78 digitalWrite(RelayPin, LOW);
79
80 delay(1000);
81 }
82
83 BLYNK_WRITE(V7) { // Replace V1 with your virtual pin number
84     int relayState1 = param.asInt(); // Get the state of the Blynk switch
85     digitalWrite(RelayPin1, relayState1);
86 }
```



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```
87  ✓ BLYNK_WRITE(V8) { // Replace V1 with your virtual pin number
88      int relayState2 = param.asInt(); // Get the state of the Blynk switch
89      digitalWrite(RelayPin2, relayState2);
90  }
91  ✓ BLYNK_WRITE(V10) { // V1 is the virtual pin for the slider in the Blynk app
92      int sliderValue = param.asInt();
93
94      // Map the slider value (0-255) to servo angles (0-180)
95      int angle = map(sliderValue, 0, 255, 0, 180);
96
97      // Set both servos to the same angle
98      servo1.write(angle);
99      servo2.write(angle);
100 }
101
102  ✓ void loop() {
103      Blynk.run();
104
105      unsigned long endTime = bme.beginReading();
106  ✓  if (endTime == 0) {
107      |      Serial.println(F("Failed to begin reading :("));
108      |      return;
109      |  }
110      Serial.print(F("Reading started at "));
111      Serial.print(millis());
112      Serial.print(F(" and will finish at "));
113      Serial.println(endTime);
114
115      Serial.println(F("You can do other work during BME680 measurement."));
116      delay(50);
117
118  ✓  if (!bme.endReading()) {
119      |      Serial.println(F("Failed to complete reading :("));
120      |      return;
121      |  }
122      Serial.print(F("Reading completed at "));
123      Serial.println(millis());
124
125      Serial.print(F("Temperature = "));
126      Serial.print(bme.temperature);
127      Serial.println(F(" *C"));
128
129      Serial.print(F("Pressure = "));
```



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```
130 Serial.print(bme.pressure / 100.0);
131 Serial.println(F(" hPa"));
132
133 Serial.print(F("Humidity = "));
134 Serial.print(bme.humidity);
135 Serial.println(F(" %"));
136
137 Serial.print(F("Gas = "));
138 Serial.print(bme.gas_resistance / 1000.0);
139 Serial.println(F(" KOhms"));
140
141 float temp = bme.temperature;
142 float pres = bme.pressure / 100.0;
143 float hum = bme.humidity;
144 float gas = bme.gas_resistance / 1000.0;
145
146 delay(1000);
147
148 float Prep = analogRead(precipita);
149 float precipitation = ((4095 - Prep) * 100 / 4095);
150 Serial.print("Precipitation: ");
151 Serial.println(precipitation);
152
153 delay(1000);
154
155 if (precipitation > 50) { // Adjust threshold as needed
156     if (!isRaining) { // Check if it's just started raining
157         isRaining = true;
158         servo1.write(0); // Close servo 1
159         servo2.write(0); // Close servo 2
160     }
161 } else {
162     isRaining = false; // Reset the rain flag
163 }
164 delay(1000);
165
166 float moist = analogRead(moisture);
167 float moistures = ((4095 - moist) * 100 / 4095);
168 Serial.print("Moisture: ");
169 Serial.println(moistures);
170
171 delay(1000);
172
```



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```
173 Serial.println();
174 float lightValue = analogRead(photoResistorPin);
175 float light = map(lightValue, 0, 4095, 0, 100);
176 Serial.print("Light: ");
177 Serial.println(light);
178 if (light < 25) {
179     digitalWrite(ledPin, HIGH);
180 } else {
181     digitalWrite(ledPin, LOW);
182 }
183
184 value = analogRead(voltageSensor);
185 vOUT = (value * 3.3) / 1024.0;
186 vIN = ((vOUT / (R2/(R1+R2)))/3.3) - 3.5;
187 Serial.print("Input Solar Voltage: ");
188 Serial.println(vIN);
189 delay(1000);
190
191 if (vIN >= 4)
192 {
193     digitalWrite(RelayPin, LOW);
194     delay(100);
195 }
196 if (vIN <= 5)
197 {
198     digitalWrite(RelayPin, HIGH);
199     delay(100);
200 }
201
202 Blynk.virtualWrite(V0, temp);
203 Blynk.virtualWrite(V1, hum);
204 Blynk.virtualWrite(V2, pres);
205 Blynk.virtualWrite(V3, gas);
206 Blynk.virtualWrite(V4, precipitation);
207 Blynk.virtualWrite(V5, moistures);
208 Blynk.virtualWrite(V6, light);
209 Blynk.virtualWrite(V9, vIN);
210
211
212 Serial.println();
213 delay(2000);
214 }
```

Figure 5.1 Coding using Arduino IDE



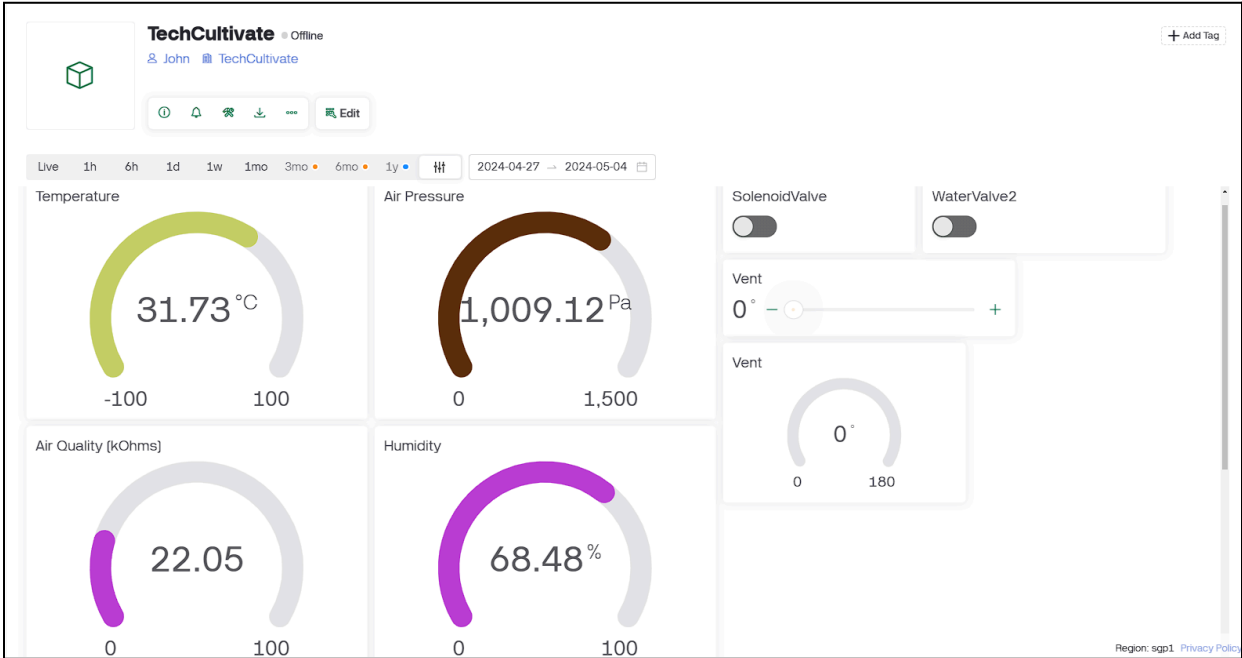
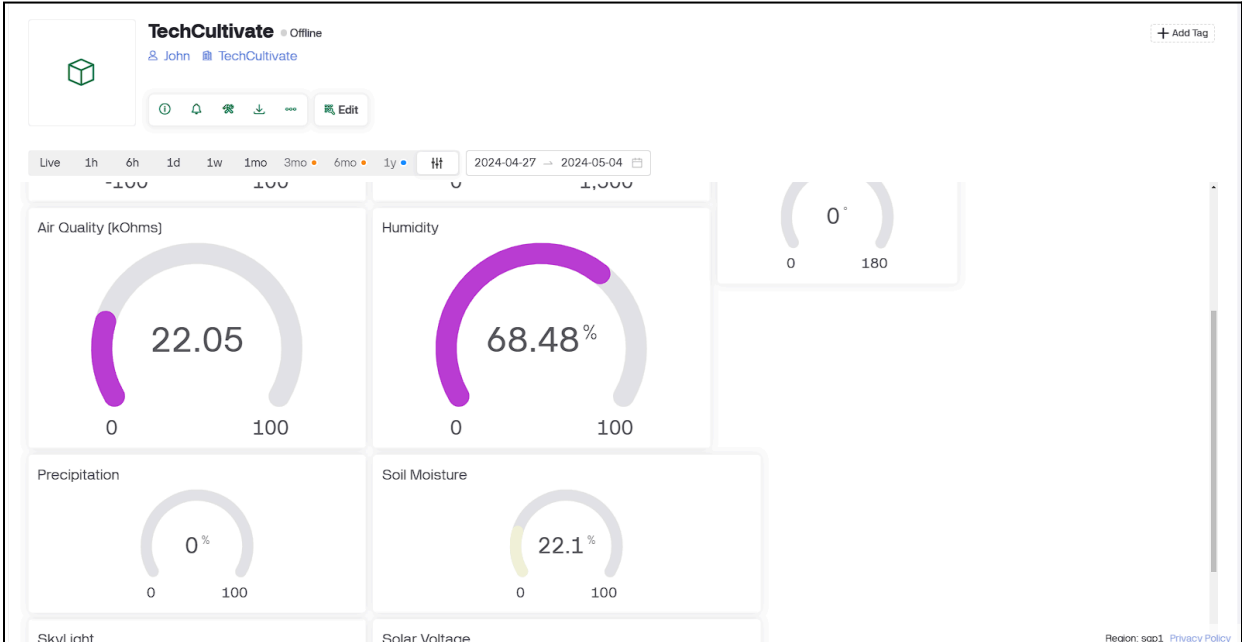
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C. BLYNK IOT





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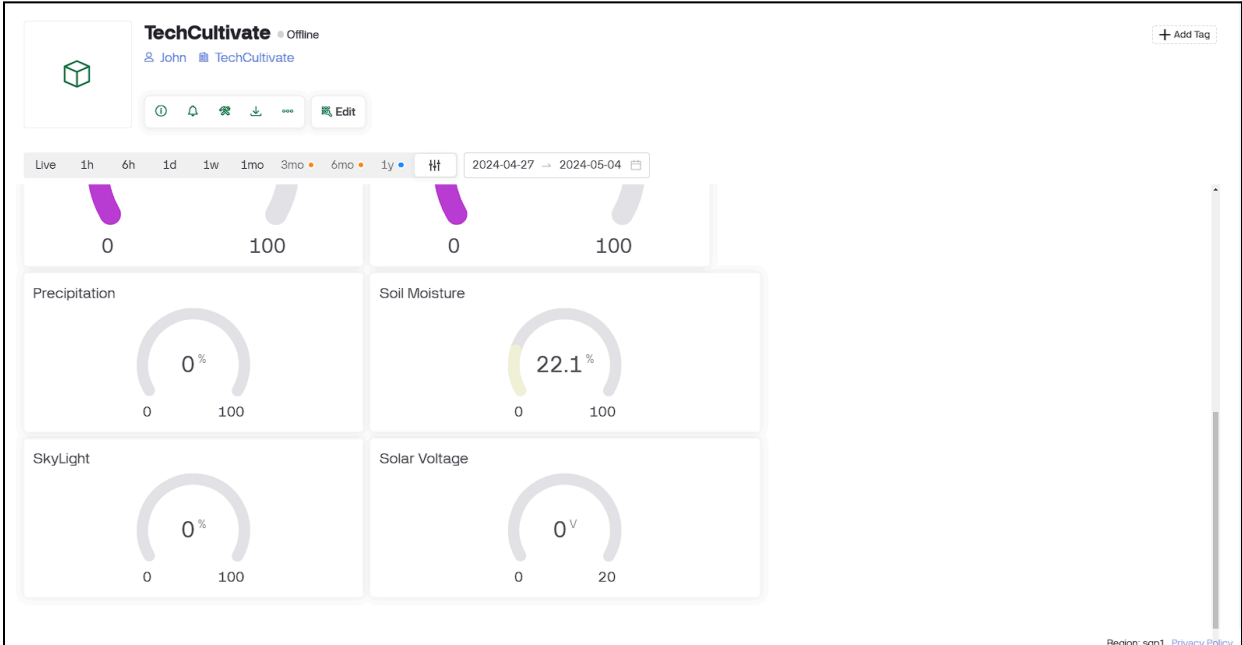


Figure 5.2 Testing Results from the Blynk IoT



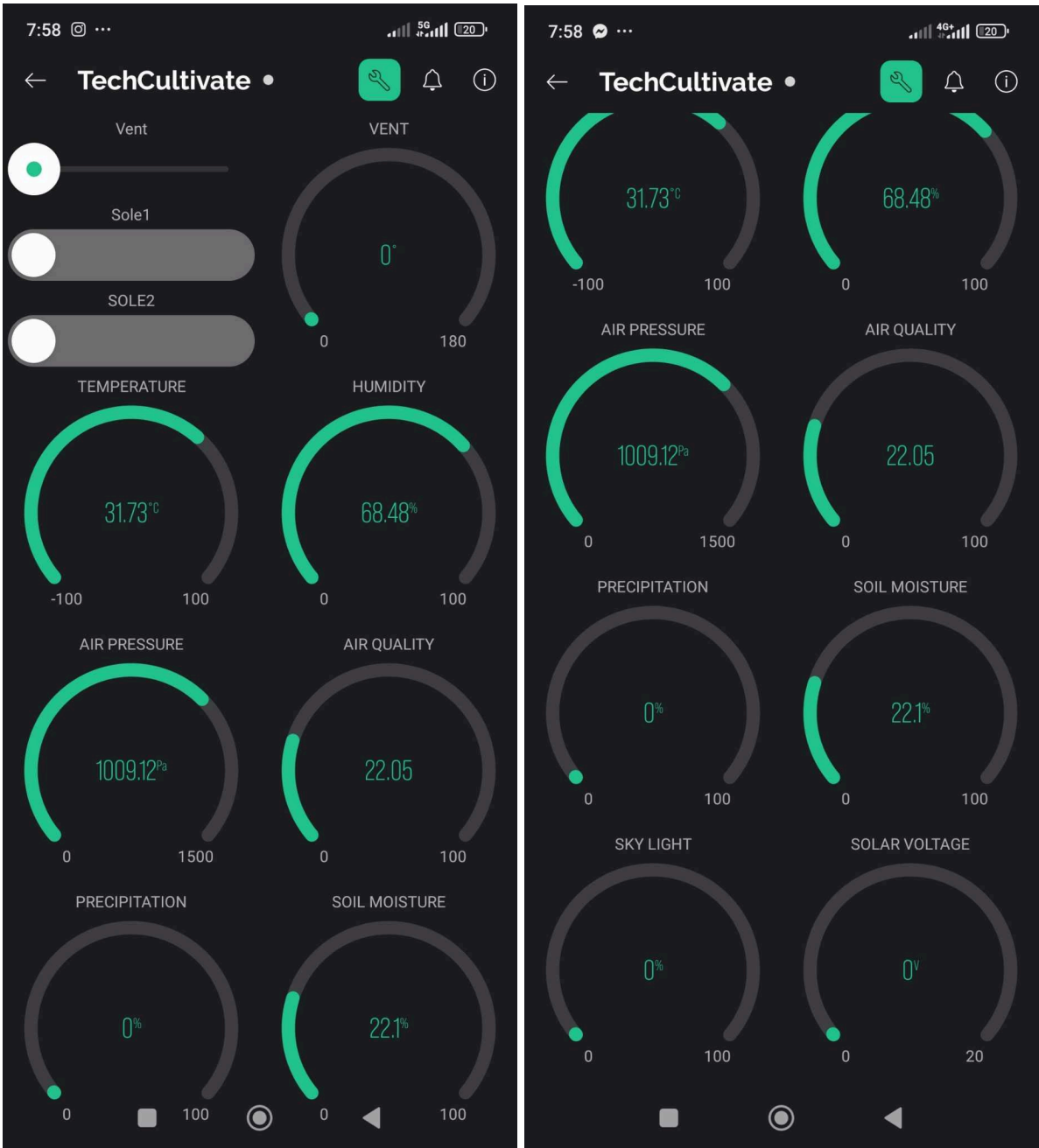
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IX. RESULTS





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Figure 5.3 Results from the Dashboard of Bkynk IoT

X. CONCLUSION

In conclusion, this overall project aims to design and create a greenhouse automated system that will reshape future greenhouse management. Through integrating different sensors and actuators, the system will be able to monitor key environmental conditions such as temperature, moisture, and rainfall. Moreover, with the incorporation of a rainfall sensor, the system will ensure optimal conditions for plant growth through automatically closing the greenhouse roof during rainfall preventing excessive water exposure and waterlogging. Furthermore, the project intends to emphasize the efficiency and safety of plant health by optimizing ventilation, the system will optimize the ventilation to prevent overheating and prioritize plant health.



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